ALEJANDRA LÓPEZ ESPINO

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ACADEMIC POSITIONS

Purdue University May 2026–present
Dean V. White Real Estate Research Fellow, Economics Department

Purdue University August 2025–May 2026
Visiting Assistant Professor, Economics Department

EDUCATION

The Pennsylvania State University *August 2015–May 2024*
Ph.D. Economics.
Dissertation Committee: Jonathan Eaton (co-chair), James Tybout (co-chair),
Marc Henry, Fernando Parro, Stephen Yeaple, and Reka Albert
Primary fields: international trade, production networks.
Secondary fields: applied econometrics and computational economics.

Instituto Tecnológico Autónomo de México (ITAM) *2015*
MA coursework in Economic Theory (66/78 credits), 2015.
BA Economics, economic theory track, 2013.

RESEARCH FIELDS

International Trade, Urban Economics, Network Formation and Analysis.

WORKING PAPERS

Production Networks and Rules of Origin: Moving from NAFTA to USMCA

Discrete Responses to Sharp Policy Shocks
with Anson Soderbery

WORK IN PROGRESS

Economic Policy and Production Network Resilience
with Chong Xiang

Digital Nomadization and Gentrification: A Case of Mexico City
with Carlos Flores Komatsu

Paved Over: the Opportunity Cost of Underutilized Parking for Affordable Housing
with Michael D. Eriksen

Effective Rules of Origin: A De Facto Restrictiveness Index for the USMCA
with José Ramón Morales-Arilla and Ernesto Stein

GRANTS AND FELLOWSHIPS

Purdue University May 2026–present
Dean V. White Real Estate Research Fellowship

Federal Reserve Bank of Chicago Summer 2021
Dissertation Fellowship

Banco de México Fall 2020
Summer Research Program

PRESENTATIONS

2026 CEPCAN Conference on Nearshoring, Global Value Chains, and the New Geography of Trade (Tec de Monterrey).

2024 Purdue University (EIIT Conference), University of Würzburg, Tübingen University (Global Economic Networks Workshop), Clemson University, Banco de México.

2021 Kiel Institute, Hitotsubashi University, Federal Reserve Bank of Chicago (Dissertation Fellowship).

2020 Banco de México (Summer Research Program).

TEACHING

Purdue University 2025–2026
Instructor

- Undergraduate international trade, and money and banking.

The Pennsylvania State University Summer 2018, 2022
Instructor

- Introduction to Econometrics, Statistical Foundations for Econometrics.

The Pennsylvania State University Fall 2015–Spring 2022
Teaching Assistant

- Undergraduate macro, micro and labor economics; master's applied microeconometrics and dissertation writing.

OTHER EXPERIENCE

The Pennsylvania State University June–December 2017
Research Assistant

- Satellite image feature extraction and large-scale estimation on PSU's HPC cluster (Prof. Michael Gechter).

ProMéxico, Secretaría de Economía May–July 2015
Consultant

Instituto Tecnológico Autónomo de México (ITAM) 2012, 2013–2015
Research Assistant

- Centro de Investigación Económica.

Secretaría de Hacienda y Crédito Público September 2012–August 2013
Research Assistant

- Tax revenue policy unit (Unidad de Política de Ingresos Tributarios).
- Public finance planning unit (Unidad de Planeación Económica de la Hacienda Pública).

TECHNICAL SKILLS

Languages	Spanish (Native), English (Fluent), French (Fluent).
Programming	R, Python, Bash, SQL, Cypher, Mathematica, Matlab, Stata.
Graph analysis	graph-tool, neo4j, NetworkX, iGraph, Cytoscape.
HPC tools	Unix/Linux package compilation (C/C++), virtual environments (micromamba), job scheduling (SLURM), large scale processing (Spark, Arrow, DuckDB, Dask), parallel processing (R, Python, Matlab).
Spatial analysis	Satellite image processing and feature extraction, vector geoprocessing (GeoPandas, Shapely, PyProj), geocoding, QGIS, ArcGIS.
AI-assisted research	Claude Code: agentic workflows for coding, data pipelines and HPC job management.
Other	Git, GitHub, VS Code, Jupyter, Hugo, LaTeX, Vim, Zotero.

PERSONAL

Citizenship	Mexico (US Permanent Resident)